

PVsyst - Simulation report

Grid-Connected System

Project: GRID CONNECTED WITHOUT SHADING

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 410 kWp

George Town - India



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PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:38
with V7.4.7

Project summary

Geographical Site

George Town

India

Situation

Latitude 13.09 °N

Longitude 80.28 °E

Altitude 24 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

George Town

Meteonorm 8.1 (1996-2015) - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Fixed plane

Tilt/Azimuth 15 / 20 °

Near Shadings

No Shadings

User's needs

Unlimited load (grid)

System information

PV Array

Nb. of modules

630 units

Pnom total

410 kWp

Inverters

Nb. of units

1 unit

Pnom total

360 kWac

Pnom ratio

1.138

Results summary

Produced Energy 635493 kWh/year Specific production 1552 kWh/kWp/year Perf. Ratio PR 82.29 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Orientation

Fixed plane

Tilt/Azimuth 15 / 20 °

Sheds configuration

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

No Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer

AE Solar

Model

AE 650ME-132

(Original PVsyst database)

Unit Nom. Power

650 Wp

Number of PV modules

630 units

Nominal (STC)

410 kWp

Modules

35 string x 18 In series

At operating cond. (50°C)

Pmpp

375 kWp

U mpp

617 V

I mpp

607 A

Total PV power

Nominal (STC)

410 kWp

Total

630 modules

Module area

1957 m²

Cell area

1834 m²

Inverter

Manufacturer

SolarMax

Model

Solarmax 360TS-SV Multi MPPT

(Original PVsyst database)

Unit Nom. Power

360 kWac

Number of inverters

1 unit

Total power

360 kWac

Operating voltage

510-800 V

Pnom ratio (DC:AC)

1.14

Power sharing within this inverter

Total inverter power

Total power

360 kWac

Number of inverters

1 unit

Pnom ratio

1.14

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const)

20.0 W/m²K

Uv (wind)

0.0 W/m²K/m/s

DC wiring losses

Global array res.

17 mΩ

Loss Fraction

1.5 % at STC

Module Quality Loss

Loss Fraction

-0.8 %

Module mismatch losses

Loss Fraction

2.0 % at MPP

Strings Mismatch loss

Loss Fraction

0.2 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000



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Main results

System Production

Produced Energy

635493 kWh/year

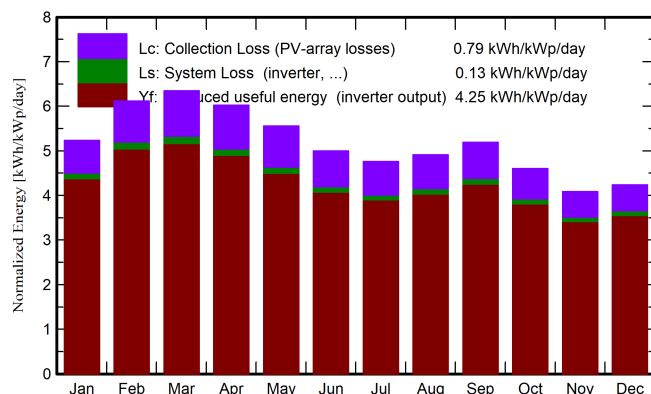
Specific production

1552 kWh/kWp/year

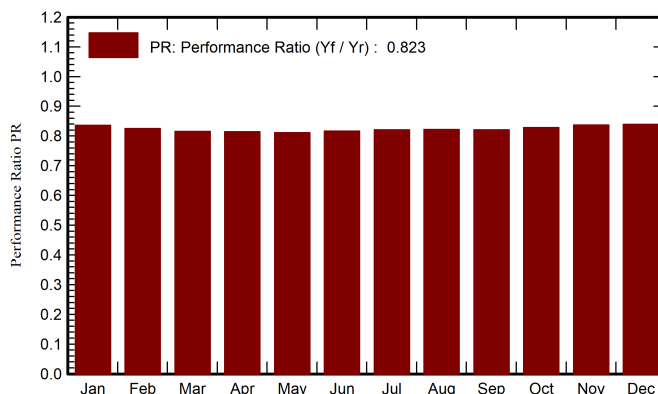
Perf. Ratio PR

82.29 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	144.2	66.32	25.11	162.4	158.9	57283	55621	0.836
February	156.4	60.47	26.21	171.4	168.0	59671	57892	0.825
March	189.7	74.76	28.44	196.7	192.8	67700	65663	0.815
April	184.7	81.64	30.16	180.7	176.8	62060	60229	0.814
May	182.6	91.33	32.17	172.3	168.1	58891	57205	0.811
June	162.3	84.49	31.21	150.0	146.1	51582	50112	0.816
July	158.0	91.51	30.64	147.6	143.6	51017	49581	0.821
August	157.1	89.73	29.68	152.3	148.7	52768	51244	0.822
September	153.7	78.09	28.85	155.7	152.2	53891	52315	0.820
October	134.9	74.42	27.86	142.8	139.9	49923	48451	0.828
November	112.4	62.00	25.92	122.7	120.0	43285	42018	0.837
December	117.8	63.02	25.21	131.4	128.5	46512	45164	0.840
Year	1853.9	917.78	28.47	1885.9	1843.6	654583	635493	0.823

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

PR Performance Ratio



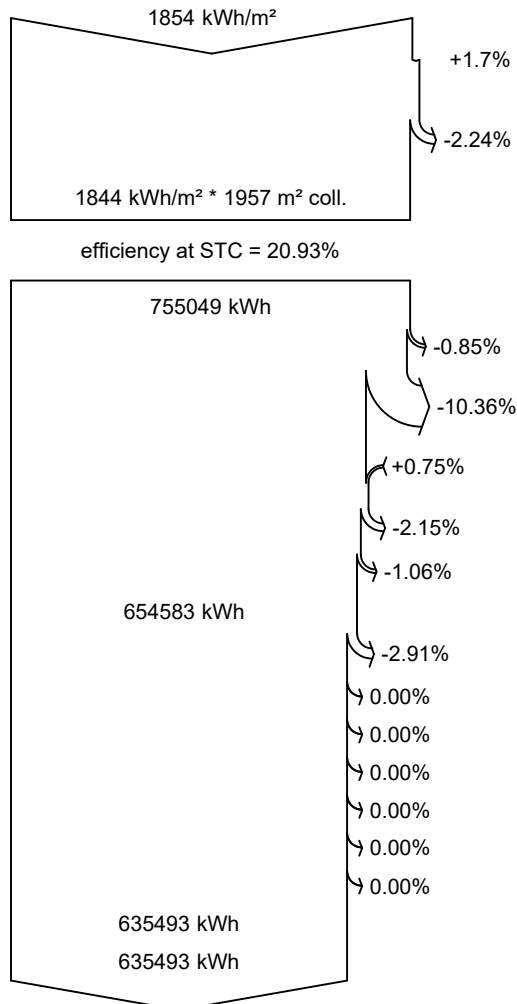
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Loss diagram



Global horizontal irradiation

Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Mismatch loss, modules and strings

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid

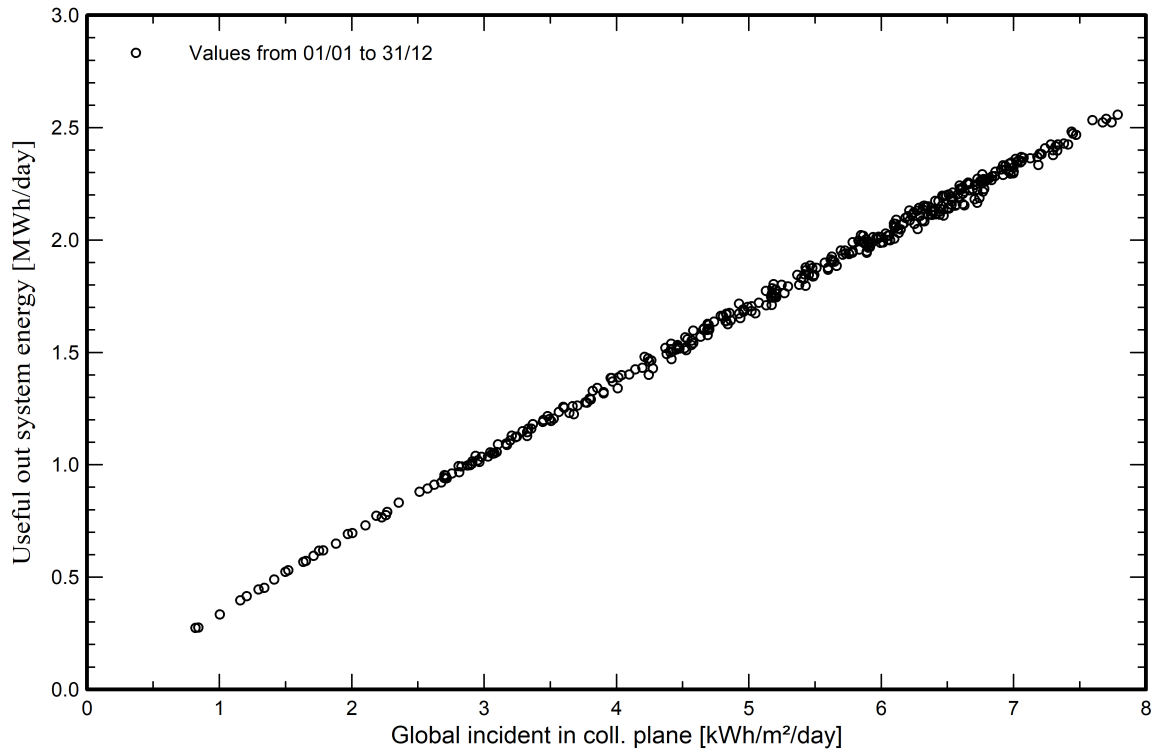


PVsyst V7.4.7

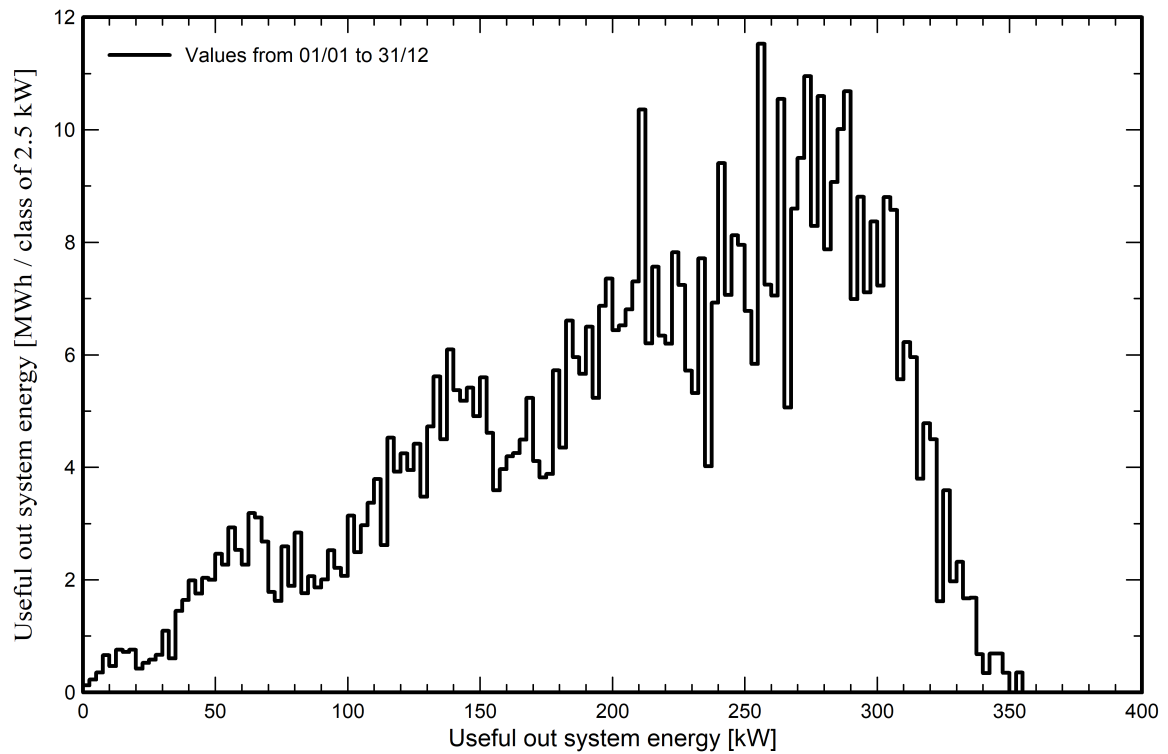
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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

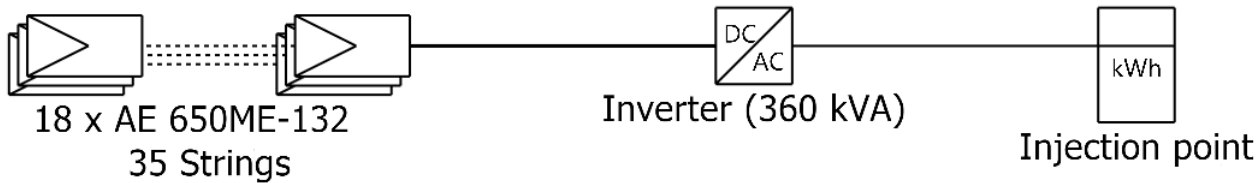




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Single-line diagram



PV module	AE 650ME-132
Inverter	Solarmax 360TS-SV Multi MPPT
String	18 x AE 650ME-132

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